

ESAT Mathematics 2 Formula + Fact Sheet

Mathematics 2 assumes all Mathematics 1 content and adds the MM topics: algebra, series, coordinate geometry, trigonometry, logarithms, calculus and graph sketching.

Exam facts: 27 multiple-choice questions, 40 minutes, no calculator. Use this sheet together with the Mathematics 1 sheet.

RATIONAL POWERS AND SURDS

$$a^m a^n = a^{m+n}, \quad \frac{a^m}{a^n} = a^{m-n}, \quad (a^m)^n = a^{mn}$$

$$a^{-n} = \frac{1}{a^n}, \quad a^{m/n} = \sqrt[n]{a^m}$$

$$\sqrt{ab} = \sqrt{a}\sqrt{b}, \quad (a + \sqrt{b})(a - \sqrt{b}) = a^2 - b$$

QUADRATIC FORMULA AND DISCRIMINANT

For $ax^2 + bx + c = 0$:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$\Delta = b^2 - 4ac$. $\Delta > 0$: two real; $\Delta = 0$: repeated; $\Delta < 0$: none.

COMPLETING THE SQUARE

$$ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 + c - \frac{b^2}{4a}$$

$$\text{Vertex: } x = -\frac{b}{2a}$$

Use for graph shape, inequalities and maximum/minimum values.

INEQUALITIES

Multiplying/dividing by a negative reverses the sign.

Quadratic inequalities: find roots and use graph/sign chart.

For rational inequalities, use critical values and a sign chart.

FUNCTIONS AND DOMAINS

One-to-one: each output has at most one input.

Many-to-one: different inputs can give the same output.

$\sqrt{x}$ means the non-negative square root.

$|x|$ is distance from 0.

SIMULTANEOUS EQUATIONS

Linear/quadratic: substitute the linear equation into the quadratic.

Use the discriminant after substitution to count intersections.

Graphically, solutions are intersection points.

POLYNOMIAL MANIPULATION

Expand brackets, collect like terms, then factorise when useful.

Algebraic division may be by linear or quadratic divisors.

Remainder degree is less than divisor degree.

COMMON ALGEBRA TRAPS

Do not cancel a term across a sum.

Check denominators are non-zero after simplifying.

After squaring an equation, substitute answers into the original.

Polynomials, sequences, series and binomial

Key extra algebra tools for Mathematics 2: polynomial division, factor/remainder theorem, recurrence, geometric series and binomial expansion.

POLYNOMIAL DIVISION

If dividing by a polynomial of degree d , the remainder has degree less than d .

For divisor $x - a$, the remainder is constant.

For divisor $ax + b$, first solve $ax + b = 0$.

FACTOR THEOREM

$f(a) = 0 \iff (x - a)$ is a factor of $f(x)$.

Use it to test roots and factor polynomials.

For factor $ax + b$, test $x = -\frac{b}{a}$.

REMAINDER THEOREM

Remainder on division by $(x - a)$ is $f(a)$.

For divisor $ax + b$, remainder is $f\left(-\frac{b}{a}\right)$.

For quadratic divisor, remainder is linear.

ARITHMETIC SERIES

$u_n = a + (n - 1)d$.

$S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$.

$1 + 2 + \dots + n = \frac{n(n + 1)}{2}$.

GEOMETRIC SERIES

$u_n = ar^{n-1}$.

$S_n = \frac{a(1 - r^n)}{1 - r}, \quad r \neq 1$.

$S_\infty = \frac{a}{1 - r}$, only if $|r| < 1$.

RECURRENCE RELATIONS

$x_{n+1} = f(x_n)$.

Generate terms carefully from the stated starting value.

Check whether indexing starts at 0 or 1.

BINOMIAL EXPANSION

$(a + b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$.

$\binom{n}{r} = \frac{n!}{r!(n-r)!} = \binom{n}{n-r}$.

For $(a + f(x))^n$, keep signs inside $f(x)$.

COEFFICIENT EXTRACTION

To find coefficient of x^k , identify the term where powers combine to x^k .

Odd powers of a negative term remain negative.

Check whether the question asks for term, coefficient, or constant term.

Coordinate geometry and circles

Straight lines, circle equations and circle properties appear explicitly in Mathematics 2.

STRAIGHT LINES

Gradient: $m = \frac{y_2 - y_1}{x_2 - x_1}$.

Point-gradient: $y - y_1 = m(x - x_1)$.

General form: $ax + by + c = 0$.

Find line equations from points, gradients or intercepts.

PARALLEL AND PERPENDICULAR

Parallel lines: $m_1 = m_2$.

Perpendicular non-vertical lines: $m_1 m_2 = -1$.

If one line is vertical, the perpendicular line is horizontal.

CIRCLE EQUATION

$$(x - a)^2 + (y - b)^2 = r^2.$$

Centre (a, b) , radius r .

Complete the square to convert from expanded form.

EXPANDED CIRCLE EQUATION

$$x^2 + y^2 + cx + dy + e = 0.$$

Centre: $\left(-\frac{c}{2}, -\frac{d}{2}\right)$.

$$r^2 = \left(\frac{c}{2}\right)^2 + \left(\frac{d}{2}\right)^2 - e.$$

TANGENTS AND CHORDS

Tangent at a point is perpendicular to the radius.

Perpendicular from centre to chord bisects the chord.

If radius is horizontal, tangent is vertical; if radius is vertical, tangent is horizontal.

CIRCLE THEOREMS

- Centre angle is twice circumference angle.
- Angle in a semicircle is 90° .
- Angles in the same segment are equal.
- Opposite angles in a cyclic quadrilateral sum to 180° .
- Alternate segment theorem.

DISTANCE AND MIDPOINT

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right).$$

Use distance for radii, chords and equal-length conditions.

INTERSECTIONS

Intersections solve simultaneous equations.

Line-circle/quadratic may give 0, 1, or 2 intersections.

Use the discriminant after substitution.

Trigonometry and radians

Mathematics 2 includes sine/cosine rules, radians, sectors, segments, trig graphs and simple trig equations.

SINE RULE

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Ambiguous SSA case: check θ and $180^\circ - \theta$, then test the angle sum.

Problems may be in 2D or 3D.

COSINE RULE

$$a^2 = b^2 + c^2 - 2bc \cos A.$$

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}.$$

Side a is opposite angle A ; angle A is between b and c .

TRIANGLE AREA

$$\text{Area} = \frac{1}{2}ab \sin C.$$

Angle C is the included angle between sides a and b .

Useful for non-right triangles.

RADIANS

$$180^\circ = \pi \text{ radians.}$$

$$\theta^\circ = \frac{\pi\theta}{180} \text{ radians.}$$

$$\theta \text{ rad} = \frac{180\theta}{\pi}^\circ.$$

Keep exact multiples of π where possible.

ARC, SECTOR, SEGMENT

For θ in radians:

$$\text{Arc length: } s = r\theta.$$

$$\text{Sector area: } A = \frac{1}{2}r^2\theta.$$

Segment area = sector area - triangle area.

TRIG IDENTITIES AND EQUATIONS

$$\tan \theta = \frac{\sin \theta}{\cos \theta}.$$

$$\sin^2 \theta + \cos^2 \theta = 1.$$

For simple equations, find principal values, use symmetry/periodicity, then restrict to interval.

TRIG GRAPHS AND PERIODICITY

$\sin x, \cos x$: period 360° or 2π .

$\tan x$: period 180° or π .

$$\sin(-x) = -\sin x, \cos(-x) = \cos x.$$

$\tan x$ is undefined where $\cos x = 0$.

EXACT VALUES

θ	0°	30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undefined

Exponentials, logarithms and graph transformations

Mathematics 2 includes log laws, exponential equations and transformed graphs. Change of base is not required.

EXPONENTIAL FORM

$$a^b = c \iff b = \log_a c.$$

$y = a^x$, for simple positive a , has horizontal asymptote $y = 0$.

If $0 < a < 1$, the graph decreases.

LOG LAWS

$$\log_a(xy) = \log_a x + \log_a y.$$

$$\log_a(x/y) = \log_a x - \log_a y.$$

$$\log_a(x^k) = k \log_a x.$$

$$\log_a(1/x) = -\log_a x, \log_a a = 1.$$

LOG DOMAINS

For $\log_a x$: $a > 0$, $a \neq 1$, and $x > 0$.

Check domain before and after using log laws.

Questions requiring change of base will not be set.

EXPONENTIAL EQUATIONS

Reduce to a common base where possible.

For $25^x - 3 \cdot 5^x + 2 = 0$, set $u = 5^x$.

After solving in u , convert back to x .

GRAPH TRANSFORMATIONS

Graph	Effect
$y = af(x)$	vertical scale $ a $; reflect in x -axis if $a < 0$
$y = f(x) + a$	up a
$y = f(x + a)$	left a
$y = f(ax)$	horizontal factor $1/ a $; reflect in y -axis if $a < 0$

COMMON GRAPHS

- Lines, quadratics and cubics.
- Trigonometric, exponential and logarithmic.
- Square root and modulus.
- Use transformations and intercepts to sketch.

QUADRATIC GRAPH FORM

$$y = a(x + b)^2 + c.$$

Vertex $(-b, c)$.

$a > 0$ opens upward; $a < 0$ opens downward.

Changing $|a|$ changes narrowness.

COMPOSITION AND LINES

$f(g(x))$ means apply g first, then f .

For domains, ensure $g(x)$ lies inside the domain of f .

Compositions of transformations may be tested.

In $y = mx + c$, m changes gradient and c changes y -intercept.

Differentiation and graph shape

Differentiation from first principles is excluded; applications to gradients, tangents, normals and stationary points are included.

DERIVATIVE MEANING

$$f'(x) = \frac{dy}{dx} = \text{gradient of tangent.}$$

Derivative can represent rate of change.

$$\text{Second derivative: } f''(x) = \frac{d^2y}{dx^2}.$$

POWER RULE

$$\frac{d}{dx} x^n = nx^{n-1}$$

Works for rational n , where the expression is defined.

Simplify roots/fractions first into powers of x .

TANGENTS

At $x = a$, tangent gradient $m = f'(a)$.

Tangent line: $y - y_1 = m(x - x_1)$.

Find $y_1 = f(a)$ before writing the line.

NORMALS

If tangent gradient is non-zero m , normal gradient is $-\frac{1}{m}$.

Horizontal tangent $\Rightarrow$ vertical normal.

Vertical tangent $\Rightarrow$ horizontal normal.

STATIONARY POINTS

Stationary point: $f'(x) = 0$.

If $f''(x) > 0$: local minimum.

If $f''(x) < 0$: local maximum.

If $f''(x) = 0$: inspect further.

INCREASING/DECREASING

$f'(x) > 0$: increasing.

$f'(x) < 0$: decreasing.

Use sign charts to test intervals.

Strictly increasing/decreasing depends on sign over the interval.

GRAPH SHAPE CHECKLIST

- Find intercepts.
- Find stationary points.
- Use derivative signs for increasing/decreasing intervals.
- Use end behaviour and asymptotes if relevant.
- Relate real roots to axis intersections.

NOT EXAMINED DIRECTLY

WATCH Differentiation from first principles is excluded.

Points of inflexion are excluded as a direct topic.

A qualitative understanding of simple polynomial inflexions is still useful.

Integration, area and trapezium rule

The distinction between a definite integral and actual area is important in Mathematics 2.

POWER INTEGRALS

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1$$

Simplify before integrating.

Applies to related sums and differences.

DEFINITE INTEGRALS

If $F'(x) = f(x)$, then:

$$\int_a^b f(x) dx = F(b) - F(a)$$

Reversing limits changes the sign.

FUNDAMENTAL THEOREM

$$\frac{d}{dx} \left(\int_a^x f(t) dt \right) = f(x)$$

Integration and differentiation are inverse operations in this sense.

COMBINING INTEGRALS

$$\int_a^b [f(x) + g(x)] dx = \int_a^b f(x) dx + \int_a^b g(x) dx$$

$$\int_a^b f(x) dx + \int_b^c f(x) dx = \int_a^c f(x) dx.$$

$$\int_b^a f(x) dx = - \int_a^b f(x) dx.$$

AREA VS SIGNED INTEGRAL

A definite integral can be negative.

Actual area between curve and axis is positive.

If the graph crosses the axis, split at roots and add absolute values.

TRAPEZIUM RULE

For spacing h :

$$A \approx \frac{h}{2} [y_0 + y_n + 2(y_1 + \dots + y_{n-1})]$$

First and last ordinates are not doubled.

DIFFERENTIAL EQUATIONS

If $\frac{dy}{dx} = f(x)$, then $y = \int f(x) dx$.

Use the given point/condition to find C .

Always include $+C$ before using the condition.

EXCLUDED STANDARD INTEGRALS

WATCH Do not assume $\int \frac{1}{x} dx$, $\int e^x dx$, or trig integrals are required.

The specified rule is for x^n , $n \neq -1$, and related sums/differences.

Mathematics 2 graph and exam checklist

Use this as the last-page checklist before timed practice: the exam is no-calculator and fast-paced.

MATHEMATICS 1 IS ASSUMED

Use all Mathematics 1 material: units, ratio, algebra, geometry, statistics and probability.
Especially useful: bounds, scale factors, graph interpretation, right-triangle trig and vectors.

AXES AND INTERSECTIONS

Axis intersections come from setting $x = 0$ or $y = 0$.

A polynomial of degree n has at most n real roots.

Intersections of two graphs solve the corresponding equations.

ALGEBRA CHECKLIST

- Could a cancelled factor be zero?
- Did divisor $ax + b$ mean substitute $-b/a$?
- Did a square root create an absolute value?
- Did a quadratic inequality need a sign chart?

GRAPH CHECKLIST

- Inside transformations reverse direction.
- Use derivatives for stationary points and shape.
- Check domain and asymptotes for log/reciprocal graphs.
- Use intercepts before sketching.

TRIG CHECKLIST

- Degrees or radians?
- All solutions in interval?
- SSA ambiguous case?
- Correct included angle in cosine rule / area formula?
- Segment = sector - triangle.

CALCULUS CHECKLIST

- Simplify before differentiating/integrating.
- Write $+C$ for indefinite integrals.
- Area may not equal signed integral.
- First/last ordinates are not doubled in trapezium rule.

NO-CALCULATOR SHORTCUTS

- Estimate first.
- Use exact forms: fractions, surds, powers.
- Eliminate impossible signs/domains.
- Check answer scale before selecting.

ATTEMPT STRATEGY

27 questions in 40 minutes $\approx$ 1.5 minutes per question.

No negative marking: eliminate, choose, flag and move.

Return to long algebra only after securing faster marks.